Gen-AI training pre-requisites and pre-read

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Overview

This document describes the setup needed to build Gen AI Apps and also gives a primer on Python to get you ready for the Gen AI training.

It is expected to take 2 to 4 hours to go through the below steps depending on your existing familiarity with these topics.

Install Python 3.12.10

1. Install Python

Install **3.12.10** version of Python from:

DO NOT INSTALL LATEST (3.13.x) PYTHON VERSION

When installing make sure the "Add python to system path" checkbox is selected

Download link:

https://www.python.org/downloads/release/python-31210/

https://www.python.org/downloads/release/python-31210/

> **Use Python 3.12.10 only as langchain and other AI python libraries do not work well with 3.13**

2. Test python

Open a command prompt and give the following command:

python –version

It should show the python version, namely, 3.12.7

If it errors finding the executable, add python.exe location to System environment variable: PATH
You can find the location of python.exe by opening the windows menu, right click on Python app (open file location), copy path

Install and setup VS Code

1. Download and install VS Code

https://code.visualstudio.com/

2. Install python extensions in VS Code

1. Open VS Code
2. Install the Python extension:
    - Click the Extensions icon in the sidebar (or press Ctrl+Shift+X)

- Search for "Python"
- Install the Microsoft Python extension

Understanding and creating a Virtual Environment for Gen AI projects

Understanding Project Root folder

Project root folder (such as **genaitraining**) or any folder in which you are putting your code, provides the base folder to work against.

When opening VS Code, this is the folder you will open (not any other subfolder)

This is important as your project virtual environment, private configuration (.env) will all reside in your project root folder.

Hence, whenever starting VS Code, always open the project root folder and not any other folder

All the code zips provided will need to be unzipped into a subfolder inside this folder

So you will eventually have a folder structure in VS Code like below:

genaitraining

|__ .env
|__ venv
|__ eda
|__ ml

..

Create a virtual environment for your project

What is Virtual Environment?

Virtual environment facilitates creation of a separate folder where all your python dependencies are saved. This ensures that you do not have to reinstall any library everytime and also that the right versions of your libraries are available to your application code (no conflicts with any other project or setup on your machine)

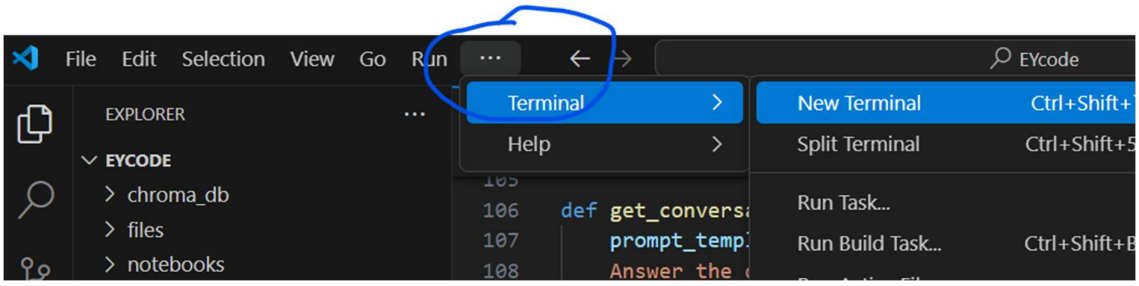
Following video explains the concept and benefits of virtual environments:

https://youtu.be/KxvKCSwlUv8?si=6HQHCeJ2wkXnJ0QX

Follow along the video below or steps outlined below to create a virtual environment

https://youtu.be/GZbeL5AcTgw?si=cvCzwnbiXPZEWFMA

1. Create a new folder for your project, called **genaitraining (or code)** in your home directory or in Desktop
2. Open the folder in VS Code (File > Open Folder)
3. Start a terminal



4. In the terminal window, give the following commands to create a new virtual environment:

   python -m venv venv

   venv\Scripts\activate

   You will see a popup asking you to use this virtual environment for your project. Click on "Yes"

Install packages needed for Gen AI development

Python package manager **pip** allows you to install all needed packages from a file commonly named as **requirements.txt**

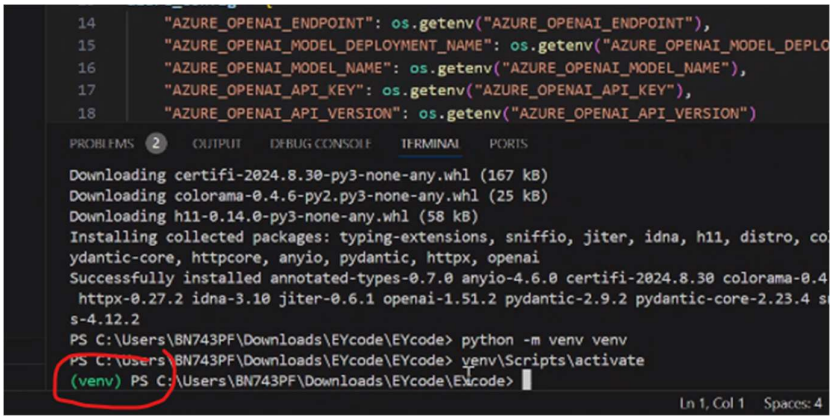
We will create requirements.txt and put all the packages we need for Gen AI development in it. We will then install those packages in our virtual environment.

1. In VS Code, with the project root folder (genaitraining or code) selected, click on the +File icon to create a new file, name it "requirements.txt"
2. Put following content in requirements.txt and save the file (ctrl+s):
   openai
   python-dotenv
   requests
   vertexai
   google-cloud-aiplatform
   langchain-openai
   langchain-community
   langchain-text-splitters
   chromadb
   beautifulsoup4

pypdf
wikipedia
plotly
tqdm
datasets
streamlit
dotted-dict
PyPDF2
matplotlib
seaborn
torch
transformers
tensorflow
tf-keras
jupyter
ipykernel

3. Install packages from requirements.txt:

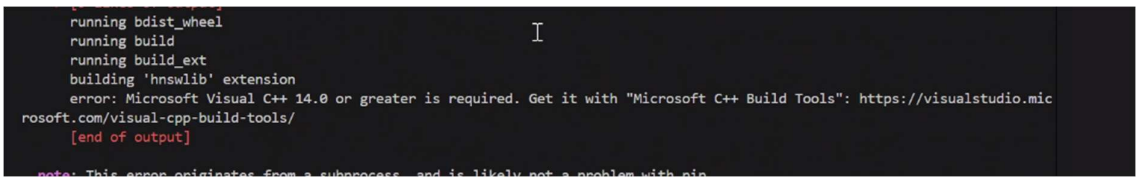
Make sure your virtual environment is active (you should see the venv name in the command prompt)



Run the following command:

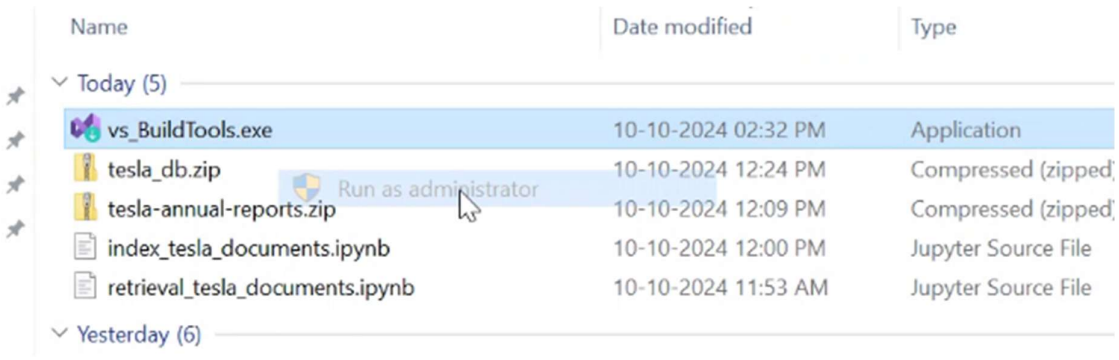
pip install -r requirements.txt

You may see error during the installation mentioning something like below:

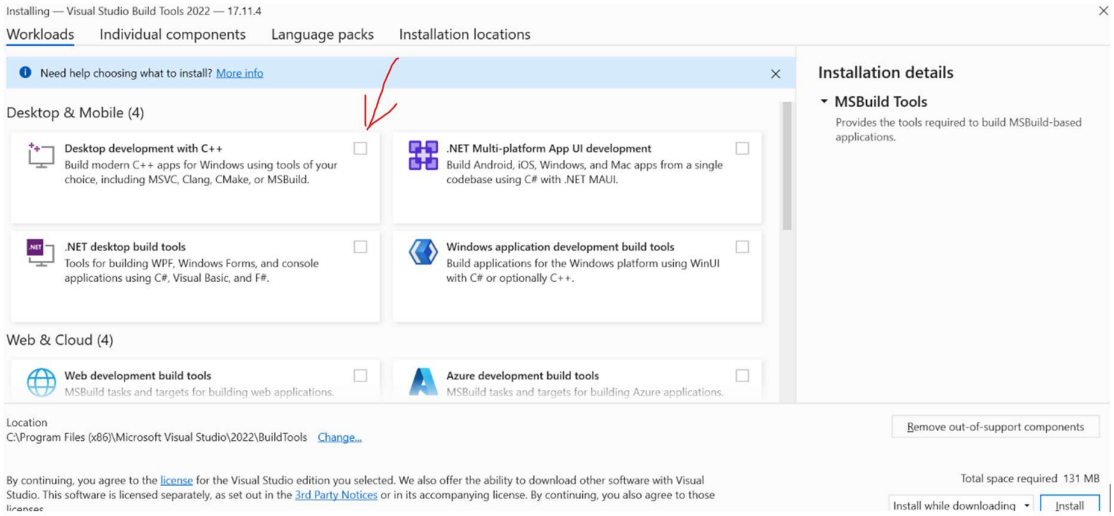


To fix it, install MS dev tools:

https://visualstudio.microsoft.com/visual-cpp-build-tools/



When you open the installer, select the first box below:



After installation, restart VS Code

And open the code folder.

Reactivate the virtual environment in the terminal:

In terminal, execute the following:

- venv\Scripts\activate

Install packages again:
pip install -r requirements.txt

Troubleshooting

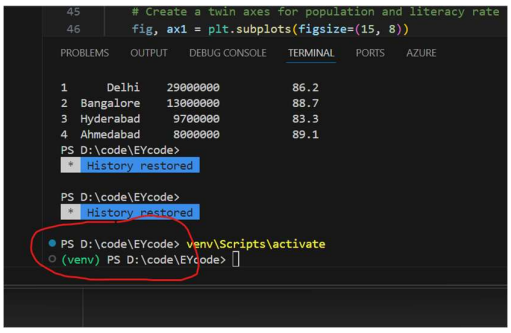
- If terminal says scripts are disabled (PowerShell):

  Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser

- If Python is not recognized, verify it's added to PATH
- Make sure to always activate virtual environment before installing packages. The active environment is shown in green as below:

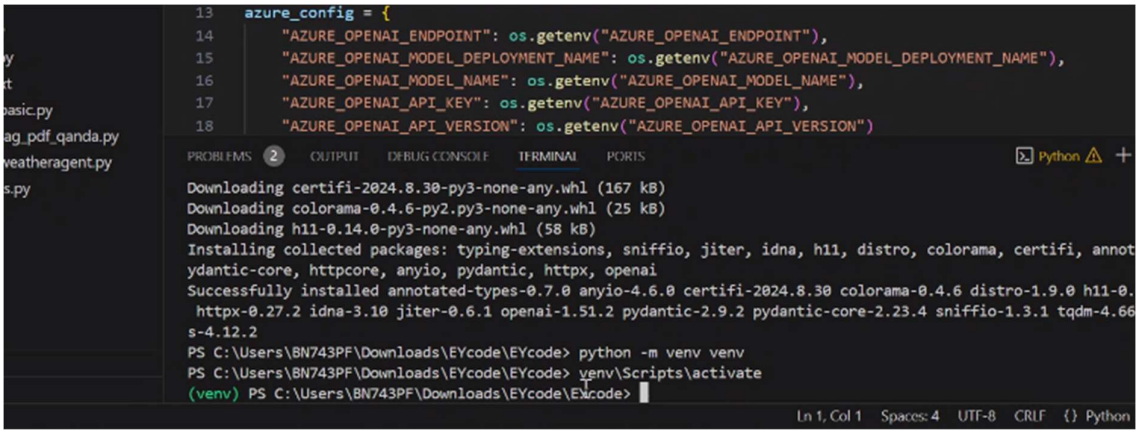
  Give command:

  venv\Scripts\activate



- If still facing errors, try following and then run *pip install -r requirements.txt*:
  pip install --upgrade
  pip setuptools wheel
  pip cache purge
  pip install --upgrade --force-reinstall setuptools

Ater successful installation of packages you should see:



Thatsit!!

You are now ready to build your GenAI Python apps with VS code

Python essentials

Download the python_basicapps.zip from:

https://github.com/vijay-agrawal/genaitraining/blob/main/code/00python_basicapps.zip

Unzip it into the **genaitraining** or your project root folder

It contains simple python scripts and a datafile **loanapproval_hist_data.csv**.
Open it in Excel and examine the data.
Loanapproved is the target column (value we want to teach the machine to predict) and other columns are features that influence the target.

Using functions in python

The demo functions_demo.py can be reviewed to understand how function calling, parameter passing is done in Python

Working with data tables using pandas

https://www.youtube.com/watch?v=tRKeLrwfUgU

Review the pandas demo code present in pandas_demo.py.

Run it by pressing the play button on top right.

Edit it to play with various dataframe capabilities such as sorting, adding a new derived column etc. Use help of your favourite AI chatbot to generate the code for the same and test it out.

Using .env for API keys and private configuration

Best practice in python based application development is to keep private configuration parameters such as API Keys, passwords etc in a **.env** file

When your python application loads it can read the private key value pairs from the .env file and use them.

This gives flexibility and security and is a common practice to deal with private configuration parameters.

See the video below on how to setup a .env file for your project:

https://www.youtube.com/watch?v=8dlQ_nDE7dQ

Open the dotenv_demo.py in VS Code

Run it by using the “Play” button on top right in your VS Code.

Alternatively, you can open the terminal, activate your virtual environment

- Venv\scripts\activate

And issue following command:

- python dotenv_demo.py

The script should run and print the environment variables present in your .env file

Working with data files

Data files such as csv files can be easily loaded in Python. They are usually converted to pandas dataframe for further analysis.

Review the file readinng demo code present in dataloading.py.

Run it by pressing the play button on top right.

You can check the data file it is loading, present in the data folder.

Using matplotlib for data visualization and analysis

Exploratory Data Analytics (EDA) with python

Run the data_visualization.py to see how matplotlib library helps with data visualization and analysis

A good video on how to do EDA with python:

https://www.youtube.com/watch?v=xi0vhXFPegw

Also search on Kaggle for existing notebooks people have made on the same.

https://www.kaggle.com/code

Accessing and consuming services using APIs

Review and run the code weather_api_json.py

It uses Weather API service from openweathermap (You can register to get your own free key)

Above should give a good handle on Python essentials for this program

Google Colab and Kaggle

Google colab and Kaggle (https://kaggle.com) provide a cloud based Python development environment using **Jupyter notebook** concept.

Notebook makes it easy to write and execute python code inline. Below are steps outlined for colab. You can do similar with Kaggle if colab is not workable for you. Kaggle also provides many pre-built notebooks by contributors that are helpful for learning.

Create your first colab notebook

Get started with Google colab by going to:

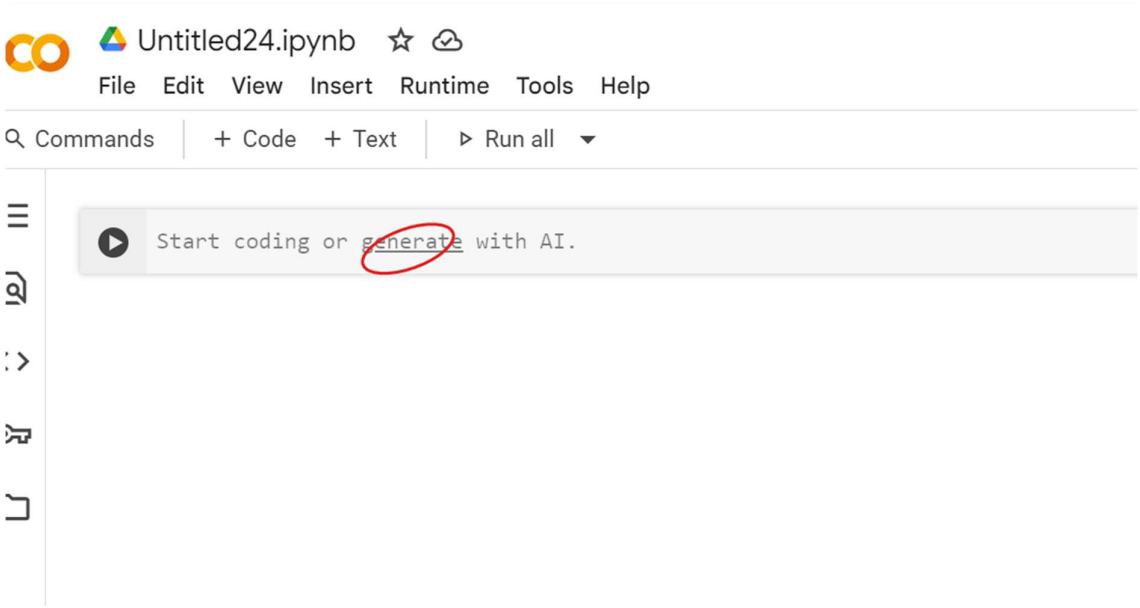
https://colab.research.google.com/ and registering with your personal gmail account.

Create your first notebook by following the video at:

https://www.youtube.com/watch?v=RLYoEyIHL6A

Try to put some of the python code from the python demos above into colab and execute them.

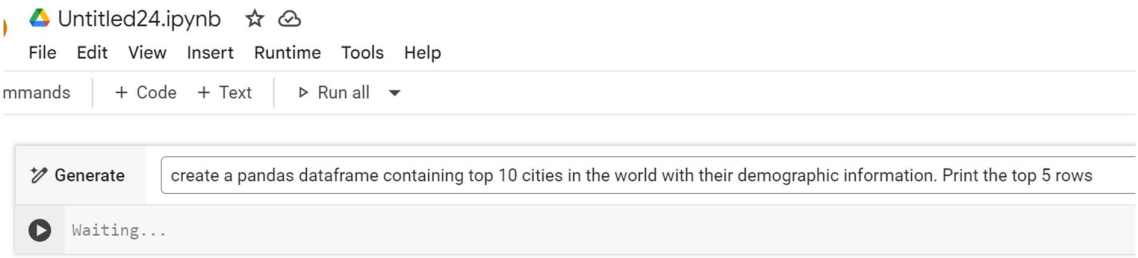
Google colab now also has **code generation** integrated into it!



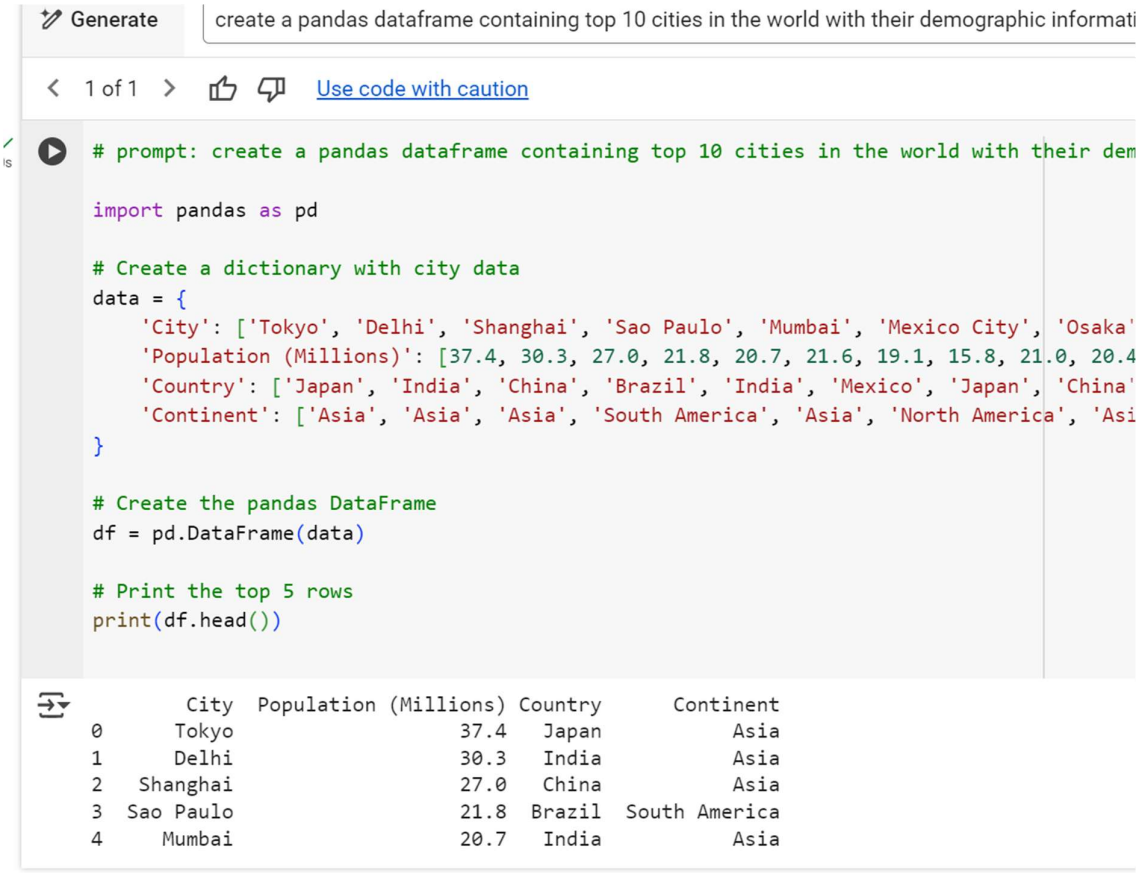
Click on “generate” and you can generate any code you want by just giving the prompt.

For example:

create a pandas dataframe containing top 10 cities in the world with their demographic information. Print the top 5 rows



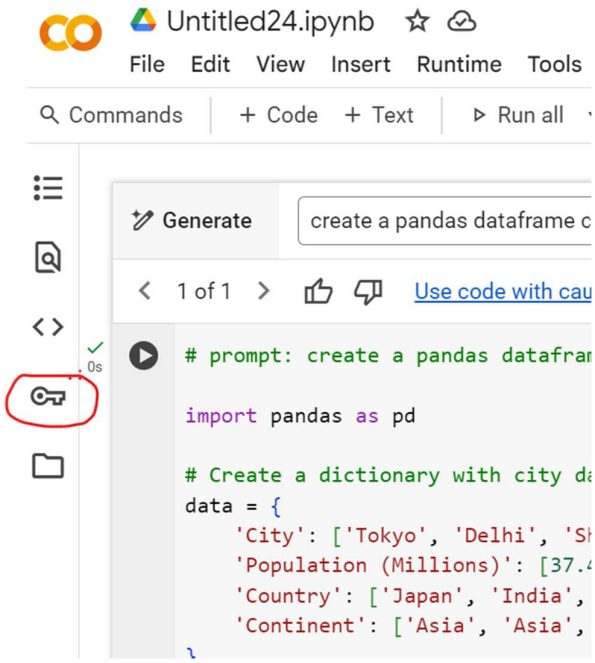
Click on the “Play” button and it will generate the code



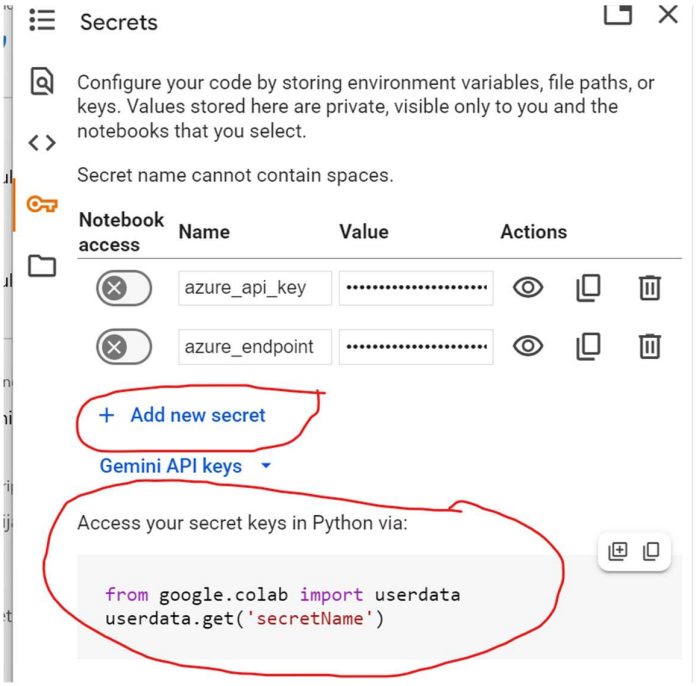
Working with secret keys in Colab

Secret Keys are a way to keep your private configuration information such as API keys secure. (It is similar to the .env concept in your python project in VS Code)

Click on the “Key” icon:



You can add new KEY just like you did in .env file



It shows the code how to use them in your notebook

Copy that code and paste it in your notebook

EXERCISE:

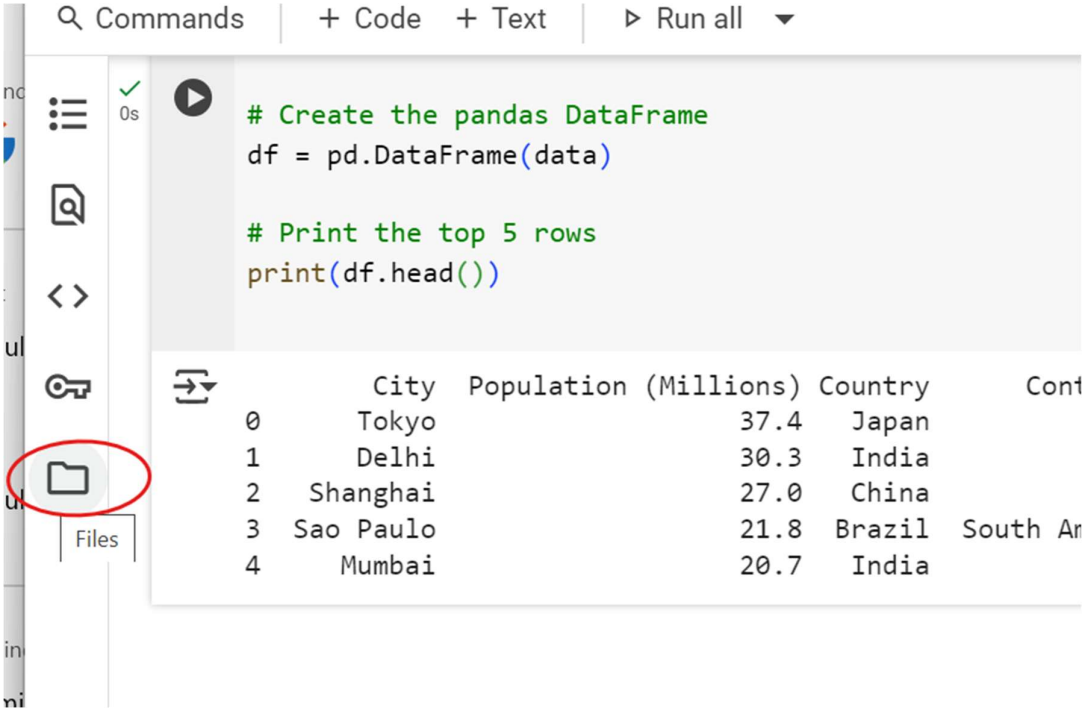
Take the code from the weather_api_json.py and make it work in colab.

Hint: remove the load_dotenv() and replace with colabs' userdata.get() after setting the OPENWEATHERMAP_API_KEY secret key in Colab

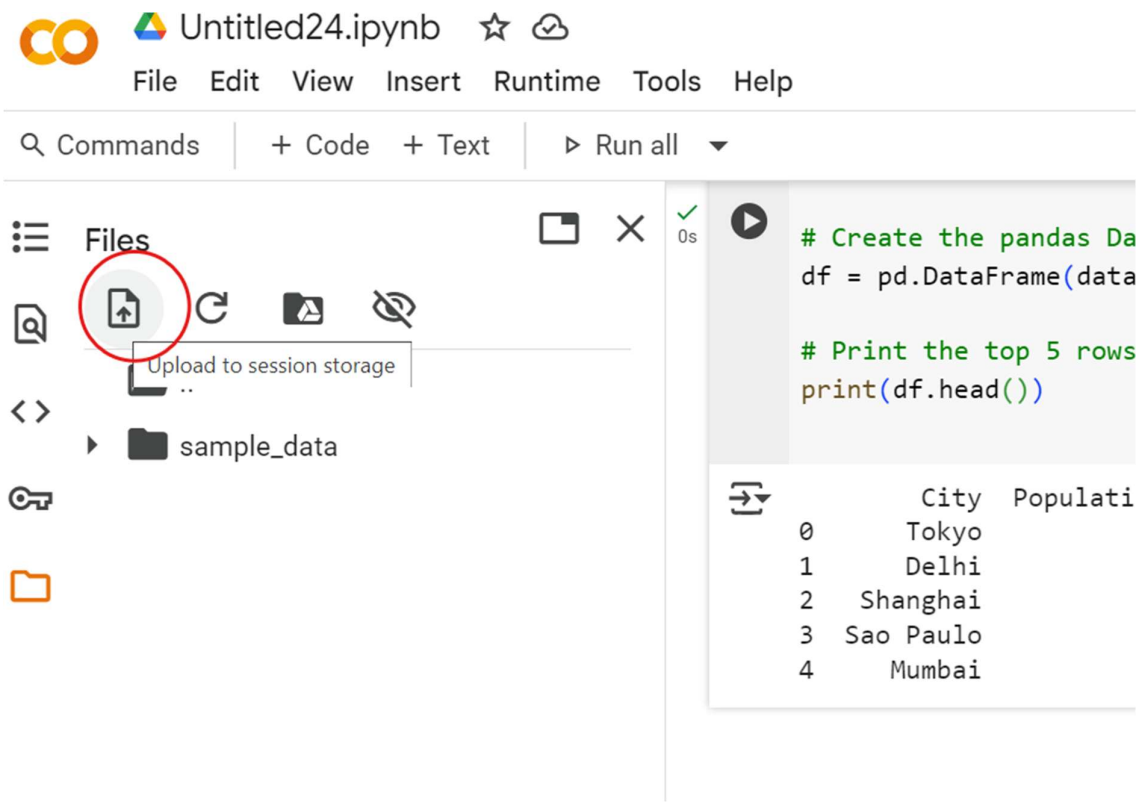
Working with data files in Colab

Colab offers a filesystem where you can upload your data files and use them in the notebook.

Click on the folder icon on the left

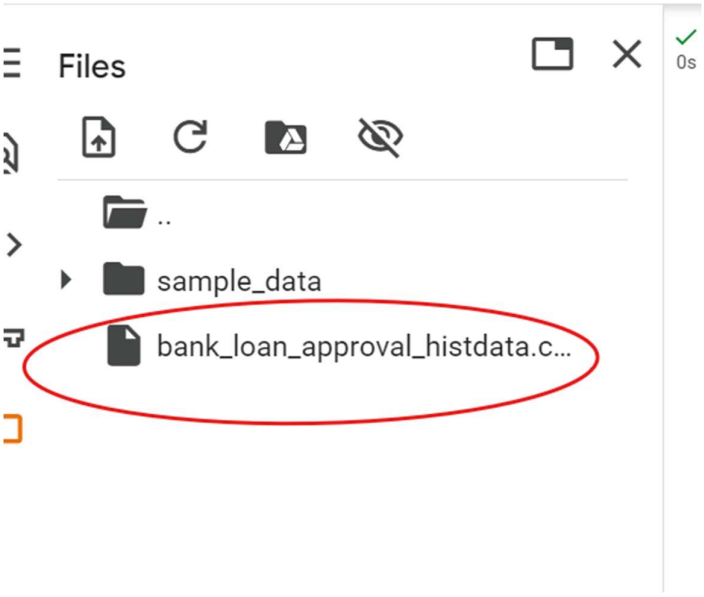


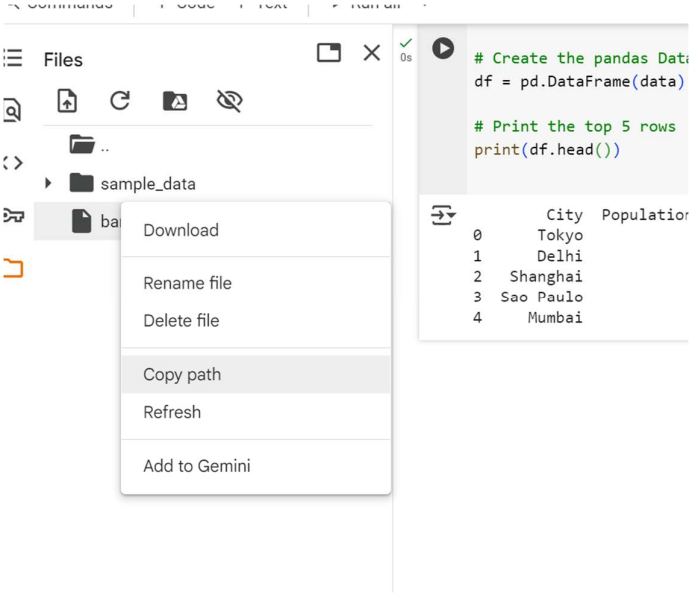
Click on the "Upload" icon to upload a data file to session storage



Once the file is uploaded it will show up in the left pane.

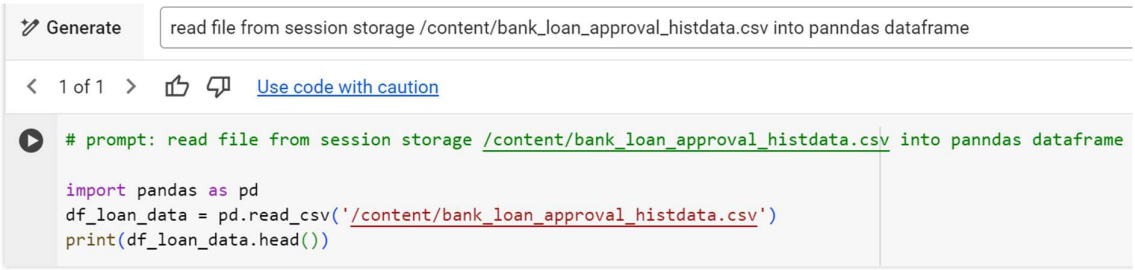
Click on the 3 dots, copy its path





Move mouse toward center of the last cell to see the +Code button. Alternatively you can click on +Code in top level menu

Generate the code as below:



When you run the cell you will see the contents of the file printed

Watch videos below for more information:

Reading csv file:

https://www.youtube.com/watch?v=VCllZKM7Njk

How to upload files and use them in colab

https://youtu.be/58t0PFIWR9Y?si=2obIrNYwyl5sk1Vq

Git (Optional)

Git is a version control tool and is needed when interacting your app with Huggingface (HF) Hugginface requires your code to be checked into a HF backed git repository

1. Download Git

1. Visit the official Git website: https://git-scm.com/download/windows

2. The download should start automatically for the latest version

3. If it doesn't start automatically, click on the download link for Windows

2. Install Git

1. Run the downloaded installer (git-X.XX.X-64-bit.exe)

2. Adjust your PATH environment:

- Select "Git from the command line and also from 3rd-party software" (recommended)

3. Complete the installation

4. Note the Path in which git is installed. You may need to add it to the System environment variables if you cannot run git from command prompt

3. Configure Git

1. Open Command Prompt or Git Bash

2. Set your username and email:
git config –global user.name “Your Name”
git config –global user.email “your email”

Note: When copying pasting the above commands, sometimes spurious characters are getting introduced. Hence manually type these commands if copy paste is not working.

4. Verify Installation

1. Open Command Prompt or Git Bash

2. Type:

git --version

3. You should see the Git version number if installation was successful

Troubleshooting

- If git commands aren't recognized, try restarting your command prompt or computer

- Ensure git is properly added to your PATH environment variable

- Check firewall settings if you have connectivity issues

For more detailed information and tutorials, visit the official Git documentation: https://git-scm.com/doc

Following videos will help understand git better:

Git overview:

https://www.youtube.com/watch?v=2ReR1YJrNOM

Git process and commands:

https://www.youtube.com/watch?v=e9lnsKot_SQ